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Validating Plume Predictions with Insights from A Decade of MMV from an Active CO₂ Injection Site

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Abstract

Dynamic model calibration is key to addressing geological complexities, ultimately providing valuable insights for optimizing storage efficiency in CCS projects. This paper aims to validate CO₂ plume predictions through a decade of monitoring, measurement, and verification (MMV) at an active CO₂ injection site in Canada. It examines the integration of geological and flow models with seismic data from the Aquistore project to refine plume predictive accuracy and enhance long-term monitoring strategies.

We used geological modeling, flow simulations, and seismic monitoring to validate CO₂ plume predictions at Aquistore. The geological model was constructed using seismic, log, and core data, incorporating key storage and sealing formations. Flow simulations were calibrated using ten years of injection data, capturing geological uncertainties. Multiple seismic surveys provided plume migration insights, with noise filtering and amplitude analysis refining detection accuracy. Dynamic model calibration incorporated reservoir heterogeneity, ensuring accurate plume representation. These findings reinforce the effectiveness of selected MMV strategies in optimizing CCS operations and establishing a benchmark for future CO₂ storage projects.

The process of analyzing seismic acquisition data for CO₂ plume detection was complex, requiring background noise exclusion and applying a defined threshold to amplitude difference maps. This method enabled the delineation of the main plume outline. This study confirms that geological heterogeneity, structural flexure, and variable reservoir properties influence CO₂ plume migration at Aquistore. Time-lapse seismic monitoring revealed that plume expansion was non-uniform across perforated intervals, reflecting differences in saturation and thickness. Refining the geological and flow models with seismic inversion data improved predictive accuracy, transitioning from a simple layer-cake assumption to one that incorporated geological complexities, such as structural flexure and reservoir heterogeneity. Observations confirm that selected MMV strategies effectively track plume behavior, demonstrating the value of 4D seismic in model validation. Calibrated model iterations strengthen confidence in CO₂ storage safety and efficacy. Ultimately, field observations and modeling results underscore the importance of integrating continuous monitoring

with adaptive modeling approaches to optimize injection strategies, minimize uncertainties, and ensure long-term CO₂ containment in saline aquifers.

This paper offers novel insights by integrating long-term CO₂ plume monitoring with advanced seismic techniques, enhancing predictive accuracy in geological storage. It advances the existing MMV methodologies through demonstrating real-time calibration of flow models. These findings play a crucial role in optimizing storage strategies, mitigating uncertainties, and improving monitoring frameworks for future CCS projects, ensuring safe and efficient long-term CO₂ sequestration.

INTRODUCTION

Background on CCS and Plume Modeling

Despite progress in renewable energy, global electricity generation remains heavily reliant on fossil fuels such as natural gas and coal ([Ember, 2025](#)). In many countries including China, India, and Canada, rising energy demand coupled with affordability keeps coal a persistent staple, underscoring the urgency of emissions mitigation. Carbon Capture and Storage (CCS) has become essential for decarbonizing high-emission sources, including power plants and industrial facilities. A world-class example is the CCS retrofit of Unit 3 at SaskPower's Boundary Dam coal-fired power plant, the world's first commercial power station with integrated CCS, capable of capturing 90% of emitted CO₂. The Aquistore project serves as the dedicated geological storage site for this initiative.

Our experience with Aquistore has shown that a key challenge in CCS deployment lies in forecasting subsurface CO₂ behavior. Accurate modeling of plume migration and containment is critical but complicated by geologic variability and structural heterogeneity. To validate these simulations, Monitoring, Measurement, and Verification (MMV) systems, including pressure sensors, fiber-optic tools, and 4D seismic imaging, are employed to track plume movement in near-real time (e.g. [Movahedzadeh et al. 2021](#)). Since 2015, the Aquistore CCS project has safely injected over 655 ktonnes of CO₂ (mid-2025) into 3.4 km-deep brine-saturated sandstone formations. Owned by SaskPower and guided by the Petroleum Technology Research Centre (PTRC), its expansive MMV framework enables continuous tracking and dynamic model refinement, laying the foundation for advanced analysis of subsurface CO₂ dynamics and calibration strategies.

Importance of MMV in CO₂ Storage

MMV is vital to CCS, ensuring containment, validating subsurface models, and meeting regulatory standards. MMV must be integrated across all CCS phases, from site selection to post-closure. A robust MMV strategy includes monitoring injection volumes, pressure, temperature, well integrity, plume movement, and chemical interactions using technologies like seismic imaging, fiber-optic sensors, and groundwater sampling ([Rostron 2017](#)). These components ensure flexible, transparent operation and adaptability. As a result, MMV can transform any CCS project from a concept into a credible, measurable climate solution by enabling real-time risk management and bolstering public trust ([Worth et al. 2014](#); [Sacuta et al. 2017](#)).

Overview of the Aquistore Project

Aquistore is the geological CO₂ storage site for SaskPower's Boundary Dam CCS initiative. Captured CO₂ has generally been distributed, primarily for enhanced oil recovery (EOR), with a smaller portion permanently stored at Aquistore ([Lee et al. 2018](#)). The site features one injection well and one observation well, drilled to depths exceeding 3.4 km and spaced 150 m apart at the surface. Both wells are constructed with CO₂-resistant materials to support integrity monitoring. The target reservoir is the Deadwood Formation, a brine-saturated sandstone in the deepest sedimentary unit of the Western Canadian Sedimentary Basin. With temperatures near 110 °C and pressures around 35 MPa, it offers optimal

conditions for deep saline CO₂ sequestration and is geologically isolated from overlying potable aquifers (Rostron et al. 2014). The area contains no known well penetrations within 25 km, except for one geothermal well. Plume behavior is tracked using seismic surveys and MMV systems. Aquistore's long-term operational record, rigorous MMV program, and favorable geology make it a benchmark for secure deep saline CO₂ storage, informing best practices in site selection, injection design, and regulatory-compliant sequestration (Figure 1).

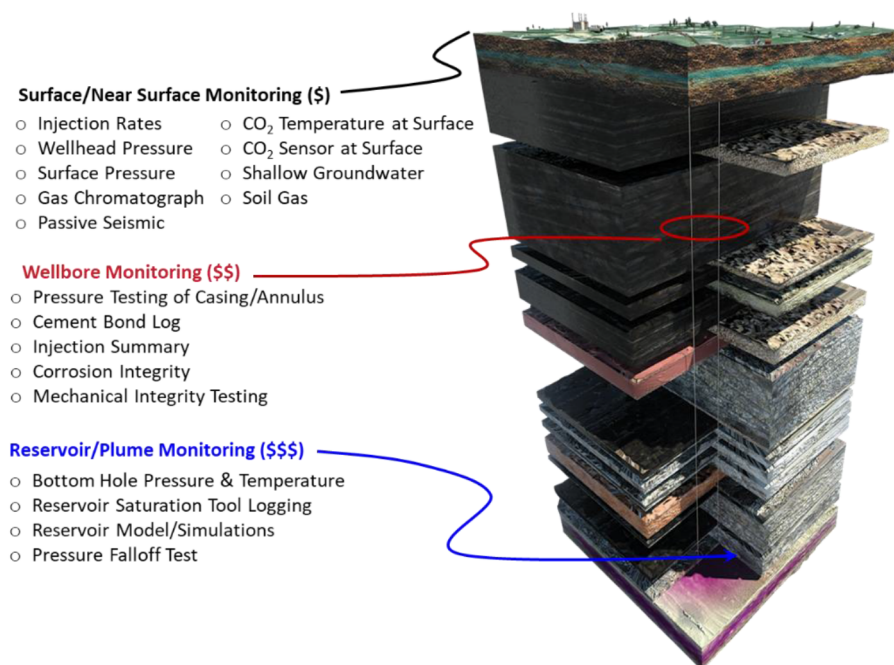


Figure 1—Visual infographic illustrating the key components of measurement, monitoring, and verification technologies at the Aquistore CCS site in Western Canada. Image courtesy of PTRC, used with permission.

Many frameworks for CCS modeling in literature still rely on simplified geology, limiting their ability to reflect complex reservoir behavior (Ajayi et al. 2019). This study evaluates the accuracy of predictive CO₂ plume models by validating them against a decade of field data from Aquistore's MMV program. By integrating geological, flow simulation, and seismic data, the study highlights strengths and gaps in current models and demonstrates the value of dynamic calibration. Analysis of plume migration and pressure evolution offers a benchmark for refining models and improving large-scale storage design. Ultimately, it shows how sustained MMV and adaptive modeling enhance containment certainty and best practices in CO₂ sequestration.

METHODOLOGY

Geological Setting, Reservoir, and Seal Formations

The Aquistore CO₂ injection site lies within a structurally complex sedimentary basin spanning southeastern Saskatchewan and Manitoba in Canada, and parts of North Dakota and Montana in the US. Regional structural elements influence depositional thickness and facies variation, but no active faulting is present near Aquistore, and hydrogeological mapping confirms long-term fluid segregation between stratigraphic units (Rostron et al. 2014). The injection zone includes two key units of the sandstone-dominated Black Island member of the Winnipeg Formation, and the Deadwood Formation overlying the crystalline Precambrian basement (Rangriz Shokri et al. 2021). Without internal seals between them, these units act as a hydraulically connected reservoir. The Precambrian basement is also perforated but contributes

minimally due to low pore space. Vertical containment is provided by the dense shale caprock of Icebox Member. Secondary seals include low-porosity carbonates and the Prairie Evaporite, whose ductile salt composition offers additional confinement. Collectively, these formations provide the petrophysical and structural characteristics necessary for long-term, secure CO₂ storage, affirming Aquistore's suitability as a deep saline sequestration site.

Construction of Geological Model

A static geological model was built for the Aquistore site to assess CO₂ storage capacity, injectivity, and containment within the basal saline system (Rangriz Shokri et al. 2024; Jiang et al., 2021). Drawing from 23 wells, the model incorporates interpreted 3D seismic data, well top correlations, and isochores to define structural layering and stratigraphy. Petrophysical properties were derived from seismic porosity volumes and refined using log and core data from 13 key wells. Detailed core-based petrophysical and geochemical analyses further informed rock behavior and CO₂-brine interactions, essential for long-term storage planning. Covering an area of 15 km × 20 km with over four million grid cells (each 72 m × 72 m), the model captures both regional structure and localized heterogeneity across formations like Deadwood, Black Island, and Precambrian basement. Sealing units, including Icebox Member shale and Prairie Evaporite salt, were explicitly modeled to evaluate containment risks and vertical migration potential. The static framework supports dynamic flow simulations calibrated against real Aquistore CO₂ injection performance, enabling reliable predictions of plume migration, pressure buildup, and temperature changes. Key parameters such as pressure regimes, injection rates, formation temperature, brine salinity, and CO₂ solubility were integrated into fluid behavior simulations (Peck et al. 2014; Torres 2017; Rangriz Shokri et al. 2019).

Injection and shut-in phase simulations support both injectivity assessment and migration forecasting. The modeling effort is closely tied to Aquistore's MMV strategy, forming a continual feedback loop between observed field data and predictive modeling. Table 1 summarizes the key model inputs used in building the geologic framework and initializing flow simulation at Aquistore.

Table 1—Summary of Flow Simulation Parameters (Adapted from Rangriz Shokri et al., 2019)

Parameters	Values
Initial reservoir pressure, kPa	34129.05 at 3173 m MD
Pressure gradient, kPa/m	10.7
Initial reservoir temperature, °C	112.8 at 3173 m MD
Initial water saturation, fraction	1
Brine salinity, ppm	300000
Porosity range, fraction	0.002-0.105
Permeability range, mD	0-55
Permeability anisotropy (k_v/k_h)	0.1
Reservoir boundary	Analytic infinite acting aquifer
Equation of State	Peng-Robinson
Default run type	Non-isothermal

Flow Simulation

A dynamic flow simulation for the Aquistore site was developed using the CMG GEM compositional simulator to replicate historical CO₂ injection. Targeting the Black Island Member and lower Deadwood Formation, the model incorporates saline aquifer properties to evaluate fluid movement and pressure evolution. CO₂ phase behavior under reservoir conditions was modeled using the Peng–Robinson equation of state, with brine, CO₂, and a trace amount of CH₄ included to stabilize numerical calculations (Rangriz

Shokri et al., 2019). CO₂ solubility was estimated via Henry's law with salinity corrections (Harvey, 1996); the initial salinity of 300,000 ppm was derived from drill stem tests across multiple Deadwood wells (Talman et al. 2020).

Reservoir pressure followed a gradient of ~10.7 kPa/m, yielding an initial pore pressure of 34.1 MPa. Temperature was set at 112 °C based on thermal logs. Drainage relative permeability curves from prior studies (Kruz et al., 2014; Bennion & Bachu, 2005) were applied to represent realistic multiphase flow across heterogeneous lithologies, including sandstone, carbonate, and shales. Spinner survey data revealed partial plugging in lower third and bottom of fourth perforation zones, which was incorporated into grid refinements and boundary conditions. The calibrated model is designed to support predictive simulation and history matching of MMV datasets, enabling accurate forecasts of plume migration, saturation development, and pressure front progression.

Aquistore's MMV Frameworks, Injection Timeline and Baseline Data

Aquistore operates one of the most advanced MMV frameworks for geological CO₂ storage, supporting multiscale observation across subsurface, surface, and atmospheric domains (e.g. Worth et al. 2014). Since initial site development, its monitoring architecture has expanded to incorporate more than two dozen technologies focused on plume tracking, anomaly detection, and containment verification within the Winnipeg–Deadwood storage complex. Surface and near-surface monitoring integrates a diverse suite of tools for comprehensive environmental surveillance. Groundwater chemistry is evaluated through piezometers installed across multiple aquifer zones, while soil gas probes assess deviations from baseline conditions (Rostron 2017; Tarakki et al. 2018). Tiltmeters and GPS monuments provide measurements of land deformation linked to subsurface pressure changes. Atmospheric CO₂ levels are tracked by sensors near the wellhead, supplemented by meteorological data from an on-site weather station. Remote sensing with InSAR imaging captures millimeter-scale elevation shifts, and annual aerial surveys are conducted to detect potential pipeline emissions or infrastructure anomalies (Czarnogorska et al. 2016).

Prior to CO₂ injection, extensive baseline investigations were conducted to confirm site suitability and guide MMV system design. During the drilling program, a comprehensive suite of open-hole logging tools, including gamma ray, spontaneous potential, resistivity, neutron and density porosity, NMR, sonic, and Modular Dynamic Testing, provided detailed insights into stratigraphy, lithology, and petrophysical properties. A mini-frac test and step-rate water injection trial established formation pressure and injectivity potential, later translated to CO₂ injection rates. Post-completion evaluations have included cement bond logs, repeated pulsed neutron decay surveys, and spinner profiling to track saturation evolution and identify zonal injectivity performance. Downhole infrastructure is extensive. The injection well is equipped with DAS/DTS fiber-optic systems, annular pressure sensors, surface wellhead gauges, and real-time injection rate analyzers. These tools collectively support continuous surveillance of injection dynamics and wellbore integrity. Located 150 meters away, the observation well features pressure and temperature sensors, fluid recovery system, bubble tube system, DTS and DAS fiber optics, and vertical seismic profile (VSP) capabilities (Ichikawa et al. 2021). These systems provide high-resolution monitoring across reservoir intervals, with repeated pulsed neutron decay logs enabling saturation tracking and conformance evaluation. Spinner logs and cement bond analysis have revealed partial perforation blockage in the lower third perforation, highlighting the role of operational diagnostics in modeling accuracy.

Seismic monitoring is conducted through a sparse permanent array of 630 wireless geophones distributed across a 12 km² grid. Co-located with the 3D seismic survey footprint, this array supports both passive and active-source imaging, including time-lapse 4D seismic acquisition and cross-well tomography. Additional coverage is provided by VSP surveys and broadband surface seismometers, delivering spectral insights into plume behavior and subsurface responses (Roach and White 2018). Aquistore's MMV technologies are categorized into multiple tiers based on function and deployment triggers including core MMV systems required for long-term compliance and performance monitoring, supplemental systems used for research-

driven characterization beyond baseline requirements, and contingency systems activated if anomalies indicate potential out-of-zone migration or deviations from modeled expectations. Table 2 provides a summary of the main MMV technologies currently deployed at Aquistore.

Table 2—Summary of Primary Measurement, Monitoring, and Verification (MMV) Technologies at the Aquistore Site

Category	System / Equipment	Function / Purpose
Operational Infrastructure	Post-Dehydration Gas Chromatograph	Analyzes CO ₂ purity and trace gases after processing
	Data Acquisition System	Remotely monitors flow rates, pressures, temperatures, and environmental parameters
	Metering Building	Houses sensors and flowmeters; integrates with control systems
Injection Well	Wellhead Pressure & Temperature Gauges	Tracks surface-level operational conditions
	Downhole Pressure & Temperature Sensors	Real-time reservoir conditions and injection performance
	Annular Pressure Sensor	Ensures casing-tubing containment integrity
	Injection Rate Sensor	Monitors instantaneous and cumulative CO ₂ injection volumes
	DAS Fiber-Optic Cable	Acoustic surveillance and seismic detection
	DTS Fiber-Optic Cable	Temperature profiling along the well
Observation Well	DAS & DTS Fiber-Optic Cable	Thermal/acoustic comparison with injection well
	Fluid Recovery System	Captures in-situ fluid samples for CO ₂ -brine reaction studies
	Bubble Tube System	Monitors formation pressure via nitrogen
	Logs (PND, spinner, VSP, cement)	assess plume distribution, flow stratification, and integrity
Seismic & Geomechanical Surveillance	Microseismic Array	Detects natural and induced seismicity for source discrimination
	Sparse Geophone Grid	630 wireless sensors for 4D plume imaging
	Broadband Surface Seismometers	Complement microseismic tools with spectral depth
	Tiltmeters	Surface tilt changes due to subsurface pressure
	GPS Gravity Monuments	Measures surface deformation linked to CO ₂ movement
	InSar	Satellite-based mapping of fine-scale elevation changes
Environmental Monitoring	Groundwater Piezometers	Checks aquifer chemistry for signs of leakage
	Soil Gas Probes	Detects abnormal gas compositions over time
	CO ₂ Air Sensors	Monitors air concentration; triggers safety shutdowns
	Aerial Pipeline Flights	Visual surveys for surface leaks along the CO ₂ transport route

MMV Adaptive Calibration Strategy

To improve predictive accuracy, the Aquistore reservoir model was iteratively calibrated using over a decade of dynamic MMV data. History matching incorporated operational metrics, such as injection volumes, bottomhole pressure, and wellbore temperature, using CMG GEM outputs. DTS fiber-optic data revealed sustained cooling from CO₂ injection, while spinner and PND logs informed flow distribution and saturation behavior. A high-resolution 5 km × 5 km model with varying grid cell size up to 4 m × 4 m around wells was aligned with the 4D seismic surveys to capture localized plume and structural dynamics. These insights were scaled up to a regional 15 km × 20 km model (4+ million cells), benchmarked against observed field responses including pressure buildup and thermal decline. Observation well data (e.g. bubble tube pressure and PND surveys) confirmed plume spread across strata. This adaptive calibration process ensured the

model's reliability for real-time MMV planning, aiding containment assurance, injection optimization, and predictive workflows for future CCS applications.

RESULTS

Model Validation through MMV Observations

Early simulation efforts at the Aquistore site successfully replicated downhole conditions, notably matching bottomhole pressure, temperature, and CO₂ injection rates over extended periods (Figure 2). However, this initial model, constructed without incorporating MMV datasets, especially 4D time-lapse seismic surveys, produced a simplified, symmetric "penny-shaped" plume that lacked geological realism.

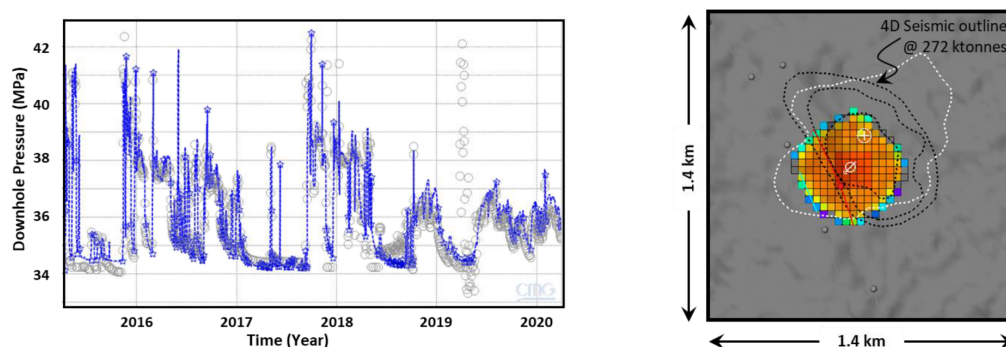


Figure 2—(a) Successful match of downhole pressure data at the Aquistore injection well using the layer-cake model assumption; (b) simulated CO₂ plume mismatch with 4D seismic survey predictions despite agreement with downhole pressure data

To address this, additional MMV inputs were introduced, including multiple rounds of time-lapse seismic imaging, well log interpretations, pulsed neutron decay (PND) logs, and bubble tube pressure measurements (Talman et al. 2022). Notably, the interpreted seismic outlines revealed an asymmetric plume configuration aligned along a North-Northwest to South-Southeast (NNW–SSE) structural flexure, an influential geologic feature reported by White et al. (2021) that appeared to restrict westward migration (Figure 3).

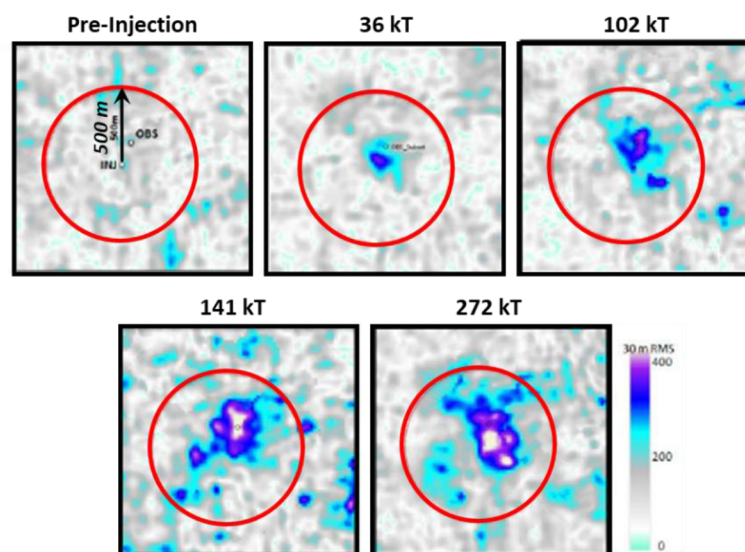


Figure 3—Top view of 4D seismic data at the Deadwood Formation showing variations corresponding to different cumulative injected CO₂ volumes as noted above each of these images (White et al. 2021)

The revised model (Figure 3) showed a dramatic improvement in plume characterization. Through iterative history matching, the model was recalibrated to simultaneously honor (i) well-based injection data, bottomhole pressure, temperature, and injection rates, (ii) plume morphology as observed in repeated 4D seismic surveys, (iii) the NNW–SSE trending flexure within the sedimentary section above Precambrian basement, and (iv) near-wellbore and field measurements including DTS fiber-optic temperature data and pressure responses at the observation well.

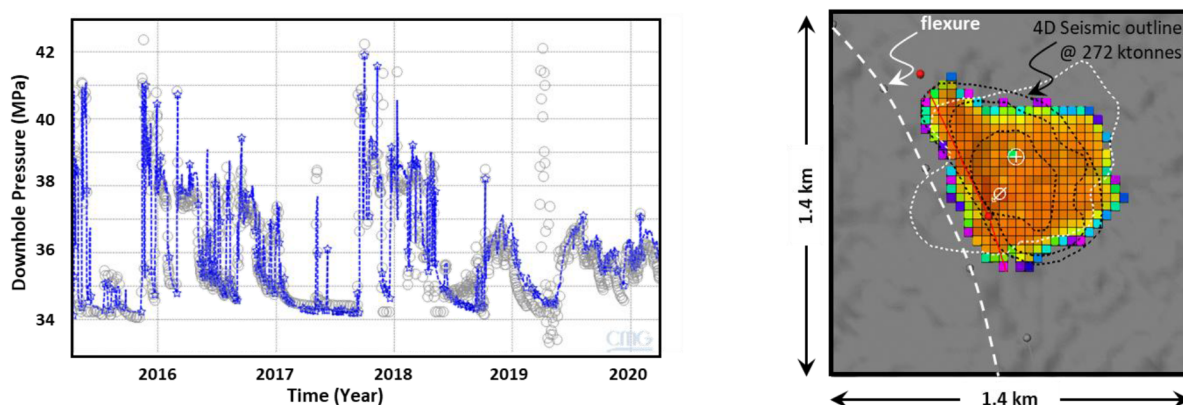


Figure 4—(a) Successful match of downhole pressure data at the Aquistore injection well using additional MMV data to constrain the model with 4D time-lapse seismic surveys and incorporation of a NNW–SSE-oriented low-permeability structural flexure; (b) simulated CO₂ plume shows improved agreement with 4D seismic survey predictions compared to results from the layer-cake model.

The revised simulation confirms that the injected CO₂ remains confined within the Deadwood–Winnipeg storage complex, with lateral migration governed by stratigraphy and lithologic heterogeneity. High-permeability sandstone beds facilitate preferential flow, while shale-rich barriers act as vertical flow restrictors. This behavior was especially evident in the second top perforated interval, where saturation levels mirror pulsed neutron log responses, validating the forward plume movement. Real-time readings from the bubble tube system confirmed hydraulic connectivity and validated reservoir compliance across multiple intervals with no evident anomalous pressure spikes or gradients indicative of leakage or vertical migration (Talman et al. 2022).

Thermal profiles, monitored via the DTS system, suggested that the cooling effect from injected CO₂ (~65–70 °C) contrasted sharply with baseline reservoir temperatures (~112 °C), offering evidence of active injection. Warming trends during shut-ins signaled thermal recovery, correlating with simulation predictions and providing further assurance of model accuracy (Rangriz Shokri et al. 2021). Again, no evidence of vertical CO₂ plume migration toward the overlying Icebox Member has been observed, reaffirming the effectiveness of the caprock as a secure containment barrier. The consistent alignment between simulated results and observed field data underscores the robustness of the current modeling approach.

Model Forecasts and Future Storage Potential

Figure 5 illustrates predicted operational injection data over a three-year period from the history-matched model, highlighted by the red shaded region, showing strong agreement with actual measured data. This alignment enhances confidence in the model's CO₂ plume prediction capabilities, bolstered by the MMV program's contributions to validation, verification, and calibration. Forward simulations based on the calibrated Aquistore reservoir model indicate stable CO₂ injection and secure long-term containment within the Winnipeg–Deadwood complex. Predicted pressure trends remain within safe operational limits, while lateral plume migration is expected to continue without compromising caprock integrity. Model outputs reinforce the site's strong hydraulic isolation and minimal leakage risk, validating its capacity for sustained

injection volumes (Vahidinia et al. 2024). Projected scale-up scenarios suggest increased injection rates can be accommodated while maintaining reservoir integrity and observing saturation thresholds critical for long-term storage.

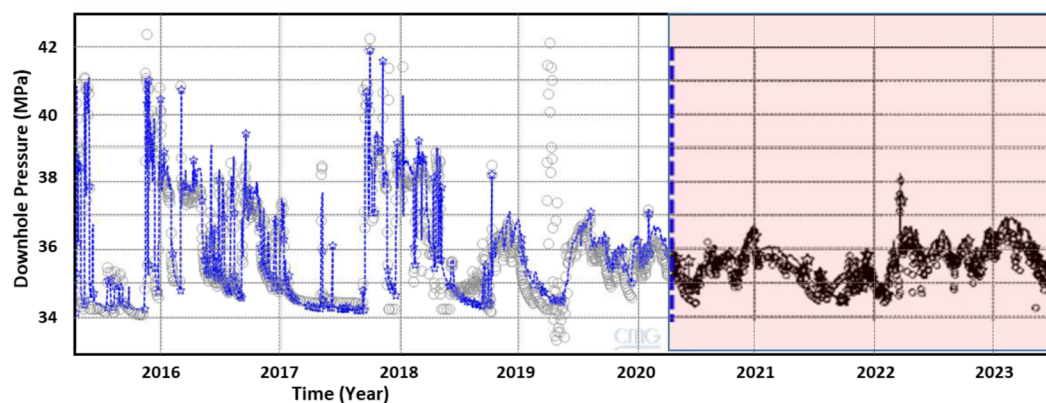


Figure 5—Predicted operational injection data over a three-year period using the history-matched model, highlighted by the red shaded region, shows strong agreement with actual measured data. This result enhances confidence in the model's capability for CO₂ plume prediction, supported by the MMV program's role in validation, verification, and calibration.

Operational flexibility in CO₂ storage remains a key strength given that the majority of CO₂ is currently utilized for EOR purposes. If additional injectivity or capacity is required, scenario-based simulations highlight the benefits of adjusting perforation zones (informed by spinner surveys and downhole camera runs), implementing intermittent shut-in phases, and leveraging MMV feedback to manage pressure buildup. Plume evolution continues to show preferential flow through high-permeability zones, ensuring consistent expansion and avoiding bypassed regions. As global carbon storage efforts accelerate, Aquistore stands as a benchmark project, demonstrating how predictive modeling and real-time MMV integration can converge to deliver secure, scalable CO₂ sequestration.

Lessons for Future CCS Projects

Aquistore presents a successful model for CO₂ storage in saline formations, leveraging a comprehensive MMV framework to achieve high-resolution surveillance with minimal surface disturbance. Seismic imaging has been essential for tracking CO₂ plume migration and validating predictive models at Aquistore (Movahedzadeh et al. 2021). High-resolution 4D time-lapse surveys, supported by VSP, cross-well imaging, and a geophone array of 630 sensors, enabled detailed visualization of subsurface plume evolution. Data interpretations consistently helped calibrate flow simulations, confirming lateral plume spread within the Deadwood and Black Island formations and containment below the Icebox Member. Seismic results were cross-validated against DTS temperature profiles, pressure logs, and pulsed neutron decay data.

The MMV-driven evolution of dynamic models, guided by field data, enables accurate forecasting and adaptive injection management. Operational flexibility and transparent reporting address challenges such as tool limitations and well remediation. Understanding subsurface heterogeneity, structural flexures, and facies variations proves vital for model validation. Early investment in seismic and petrophysical analysis, and other monitoring technologies (e.g. bubble tube) reinforces confidence in long-term storage security. Ultimately, this integrated MMV approach improves model accuracy, confirms containment, and positions seismic imaging as a cornerstone of monitoring and modeling validation.

CONCLUSIONS AND RECOMMENDATIONS

The Aquistore CO₂ storage project provides a technically validated and operationally proven case study of deep saline injection. Through the integration of static geologic modeling, high-resolution compositional

simulations, and an expansive suite of MMV tools, the project has achieved demonstrable containment, conformance, and predictability of CO₂ behavior over nearly a decade of injection. History-matched models closely align with field performance, validating plume migration patterns, pressure evolution, and saturation changes across the Winnipeg-Deadwood storage complex. Field data from pressure sensors, pulsed neutron logs, DTS systems, and seismic imaging confirm that CO₂ remains within modeled bounds and that caprock integrity is preserved.

Key insights from model calibration emphasize the role of stratigraphic heterogeneity and operational anomalies (i.e. intermittent injection) in influencing reservoir performance. Geological controls, including a structural flexure pattern and facies variation, have modulated flow behavior, underscoring the necessity of integrated seismic and petrophysical analysis during site selection and model construction. The Aquistore MMV architecture, featuring co-located surface, well, and subsurface sensors, demonstrates how layered monitoring can enable real-time validation and adaptive reservoir management. As CCS efforts expand globally, Aquistore provides a scalable MMV framework to support scientific advancement, policy and regulatory decision-making processes, and public engagement in carbon sequestration.

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